[redacted]

Alright 😄 this is **peak Kubernetes philosophy + meme material**

> *"Who schedules the scheduler?"*
> Yes, welcome to the **Chicken-Neck Problem** 🐔🦙 (Kubernetes edition).

I'll clean this up, keep your flow, add structure, and sprinkle senior-engineer clarity + mild humor.

🧩 Static Pods – *"Who watches the Watchmen?"*

Step 0 – The Way → Problem → Solution (Big Picture)

🛣️ The Normal Way

- You create a Pod
- **kube-scheduler** assigns it to a node
- kubelet runs it

  ✅ Smooth pipeline.

💥 The Problem (Mind-bender)

You discover:

- **Scheduler itself is a Pod**
- It runs in `kube-system` namespace

```bash
kubectl get pods -n kube-system | grep scheduler
```

Now brain explodes 🤯
👉 **Who schedules the scheduler?**

If *every Pod needs a scheduler*
and
the scheduler is also a Pod

🚨 **Chicken-neck problem unlocked**

🛠️ The Solution

💡 STATIC PODS

They are:

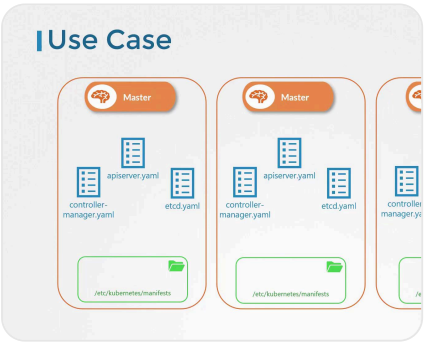
- Not scheduled by kube-scheduler ❌
- Directly managed by **kubelet** ✅

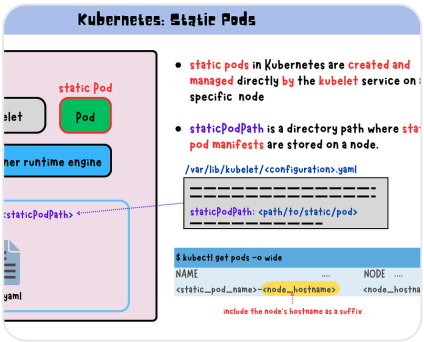
Step 1 – Overview (High-level understanding)

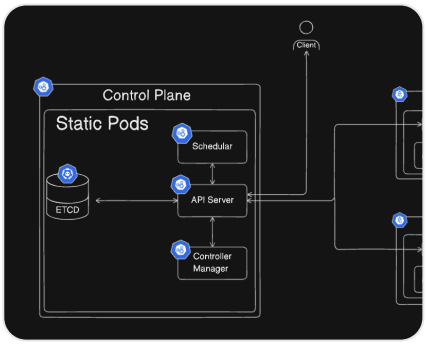
Static Pods are:

- Pods defined **directly on a node**
- Created from YAML files on disk
- **kubelet watches a directory**
- If YAML appears → Pod runs
- If YAML disappears → Pod dies ☠️

📌 No scheduler involved
📌 No API request needed
📌 Node-local truth







Step 2 – Core Concepts (Let's untangle the logic)

🔑 Key Truths

1️⃣ All control plane components are Pods

- kube-apiserver
- kube-scheduler
- kube-controller-manager

- etcd

They live in:

```bash
kube-system namespace
```

2️⃣ kubelet exists on EVERY node

Not just worker nodes 😎

```bash
kubectl get nodes
```

Control-plane is ALSO a node.

3️⃣ Static Pods are managed by kubelet

- kubelet scans:

  ```bash
  /etc/kubernetes/manifests/
  ```

- Finds YAML → runs Pod
- Deletes YAML → deletes Pod

🧠 kubelet = **local Pod manager + babysitter**

Step 3 – Hands-on Example (Real proof, no magic 🪄)

🔍 Verify scheduler Pod

```bash
kubectl get pods -n kube-system -o wide | grep scheduler
```

Example:

```text
kube-scheduler-cka-cluster1-control-plane
NODE: cka-cluster1-control-plane
```

👉 Scheduler is running **on control-plane node**

🧱 Enter control-plane node (Kind / Docker case)

```bash
docker ps | grep control-plane
docker exec -it cka-cluster1-control-plane bash
```

Now you are **inside control-plane** 🕶️

🔎 kubelet is running

```bash
ps -ef | grep kubelet
```

📂 Static Pod directory

```bash
cd /etc/kubernetes/manifests/
ls -lrt
```

You'll see:

```text
etcd.yaml
kube-apiserver.yaml
kube-controller-manager.yaml
kube-scheduler.yaml
```

🎯 THESE ARE STATIC PODS

💣 Delete the Scheduler (controlled chaos)

```bash
mv kube-scheduler.yaml /tmp/
```

👀 Check from another terminal

```bash
kubectl get pods -n kube-system
```

❌ Scheduler pod is GONE

🧪 Create a new Pod

```bash
kubectl run nginx-pod --image=nginx
kubectl get pods
```

Result:

```text
nginx-pod   Pending
```

👉 Why?
❌ No scheduler
❌ No node assignment

Kubernetes did **NOT crash**
It just politely waited 🧘

🔄 Bring Scheduler back

```bash
```

```
mv /tmp/kube-scheduler.yaml /etc/kubernetes/manifests/
```

📌 kubelet notices immediately
📌 Scheduler pod starts again

✅ Final check

```bash
kubectl get pods
```

```text
nginx-pod   Running
```

🎉 Pending → Running
The king has returned 👑

Step 4 – Common Mistakes (Don't shoot your cluster 😅)

❌ Editing static pod YAML with syntax errors
➡️ Component won't start

❌ Forgetting control-plane is also a node
➡️ Confusion during troubleshooting

❌ Thinking static pods are visible like normal pods
➡️ They **are**, but lifecycle is node-local

❌ Restarting control-plane Pods using `kubectl delete`
➡️ kubelet will resurrect them anyway 😈

Step 5 – Advanced Usage / Optimization (Senior stuff 🧠)

🛠️ Why Static Pods are used for control-plane?

- Scheduler dependency problem solved
- Faster bootstrap

- Works without scheduler
- Works without controller-manager

🔄 How do we “restart” control-plane components?

👉 Edit files here:

```bash
/etc/kubernetes/manifests/
```

Change = restart

Remove = stop

Add = start

No systemctl. No drama.

🧠 One-Line Memory Trick (Interview Gold)

> **“Normal Pods need scheduler.**
> **Scheduler needs kubelet.**
> **kubelet reads static YAML.”**

If you want next:

- Static Pods vs DaemonSets
- How kubeadm uses static pods
- Debugging broken control-plane (CKA-style)
- What happens if etcd static pod dies 😈

Say the word 😄